



24CSA449_Artificial Intelligence

Lecture I: Introduction to AI



Overview



Introduction to AI



AI-History and evolution



Key concepts and definitions



Overview of Machine learning



Neural networks,



Natural language processing

A futuristic robot with a glowing blue eye and a city skyline in the background. The robot is white and metallic, with a large, circular, glowing blue eye. It is standing in front of a city skyline at night, with a globe and a network of glowing blue lines in the foreground. The robot's hand is raised to its chin in a thinking pose.

Introduction to AI

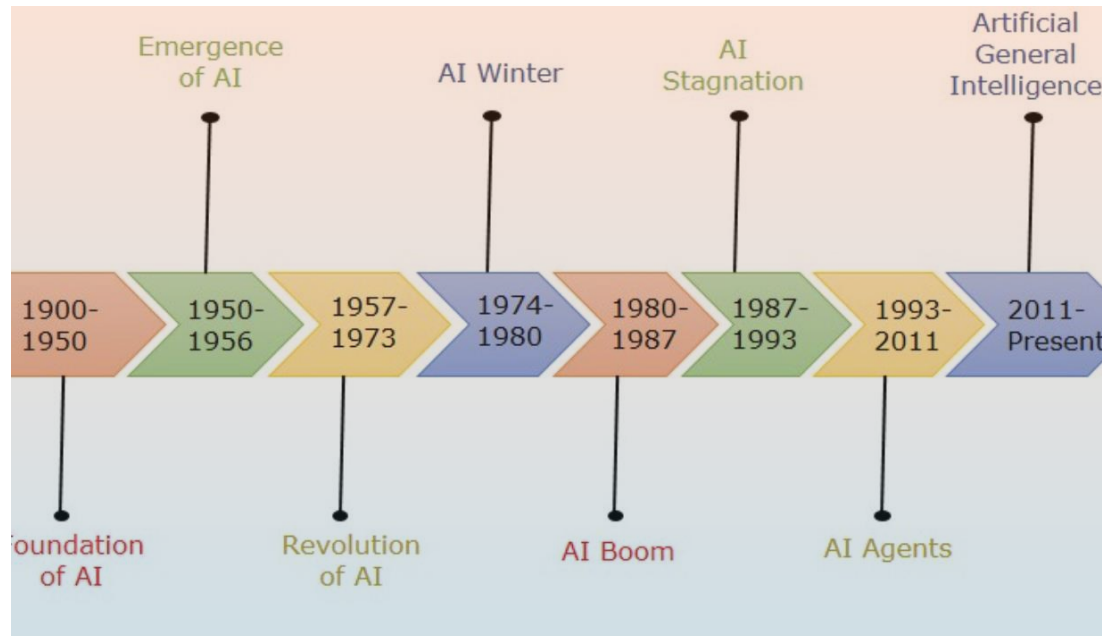
- **What is AI?**
 - **The power of a machine to copy intelligent human behavior.**
 - **Study of how to make computers do things, which at the moment, people do better.**
 - **Simulate human cognitive functions and exhibit autonomous behavior.**

Introduction...

- Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that typically require human intelligence.
- These tasks include learning, reasoning, problem-solving, perception, understanding natural language, and decision-making.
- AI aims to develop machines that can simulate human cognitive functions (Human cognitive functions refer to the mental processes by which humans acquire knowledge and understanding like perception, attention, memory, learning, problem solving, decision making etc) and exhibit autonomous behavior.



History and Evolution



- + • 1940s–1950s: Foundations of AI; early research on intelligent machines.
- 1950: Alan Turing proposed the Turing Test to evaluate machine intelligence.
- • 1956: Dartmouth Conference coined the term Artificial Intelligence (birth of AI).
- 1956–1970s: Development of early AI programs for problem-solving, games, and reasoning.
- 1970s–1980s: AI Winter due to limited computing power and reduced funding.
- 1980s: Rise of Expert Systems for medical, financial, and industrial applications.
- 1990s–2010s: Growth of Machine Learning using data-driven approaches.
- 1997: IBM Deep Blue defeated world chess champion Garry Kasparov.
- 2012: Deep Learning breakthrough significantly improved image and speech recognition.
- 2016: AlphaGo defeated Go champion Lee Sedol.
- 2020s–Present: Rise of Generative AI and Large Language Models (LLMs) for text, images, code, and multimodal applications.

Need for A.I

- **Efficiency and Automation**
 - **Task Automation:** AI can automate repetitive tasks, reducing the need for human intervention. This is particularly useful in manufacturing, customer service, and administrative tasks.
 - **Process Optimization:** AI algorithms can optimize business processes, reducing costs and improving efficiency
- **Data Analysis:**
 - **Big Data:** AI can analyze vast amounts of data quickly and accurately, providing insights that would be impossible for humans to glean in a reasonable timeframe
 - **Predictive Analytics:** AI can predict trends and behaviors by analyzing historical data, helping businesses make informed decisions.
- **Personalization:**
 - **Customer Experience:** AI can provide personalized recommendations and experiences in real time, enhancing customer satisfaction. Examples include recommendation systems used by Netflix and Amazon
 - **Healthcare:** AI can personalize treatment plans based on individual patient data, improving health outcomes.
- **Innovation and Problem-Solving:**
 - **New Solutions:** AI can solve complex problems that were previously unsolvable, leading to innovations in various fields such as medicine, climate science, and engineering
 - **Research:** AI accelerates research and development by analyzing scientific data, generating hypotheses, and even conducting experiments.

Need for A.I

- **Safety and Security:**
 - **Risk Management:** AI systems can detect and respond to security threats faster than humans, enhancing cybersecurity
 - **Health and Safety:** AI-powered monitoring systems can improve safety in industries like construction, mining, and healthcare.
- **Improved Decision-Making:**
 - **Business Intelligence:** AI provides actionable insights and forecasts that aid in strategic decision-making
 - **Real-Time Responses:** AI systems can analyze and respond to data in real time, crucial for applications like stock trading, autonomous driving, and emergency response.
- **Accessibility:**
 - **Assistive Technologies:** AI can help individuals with disabilities by providing tools such as speech recognition, real-time translation, and automated text-to-speech
 - **Language Translation:** AI-powered translation services break down language barriers, making information more accessible globally



Characteristics of A.I

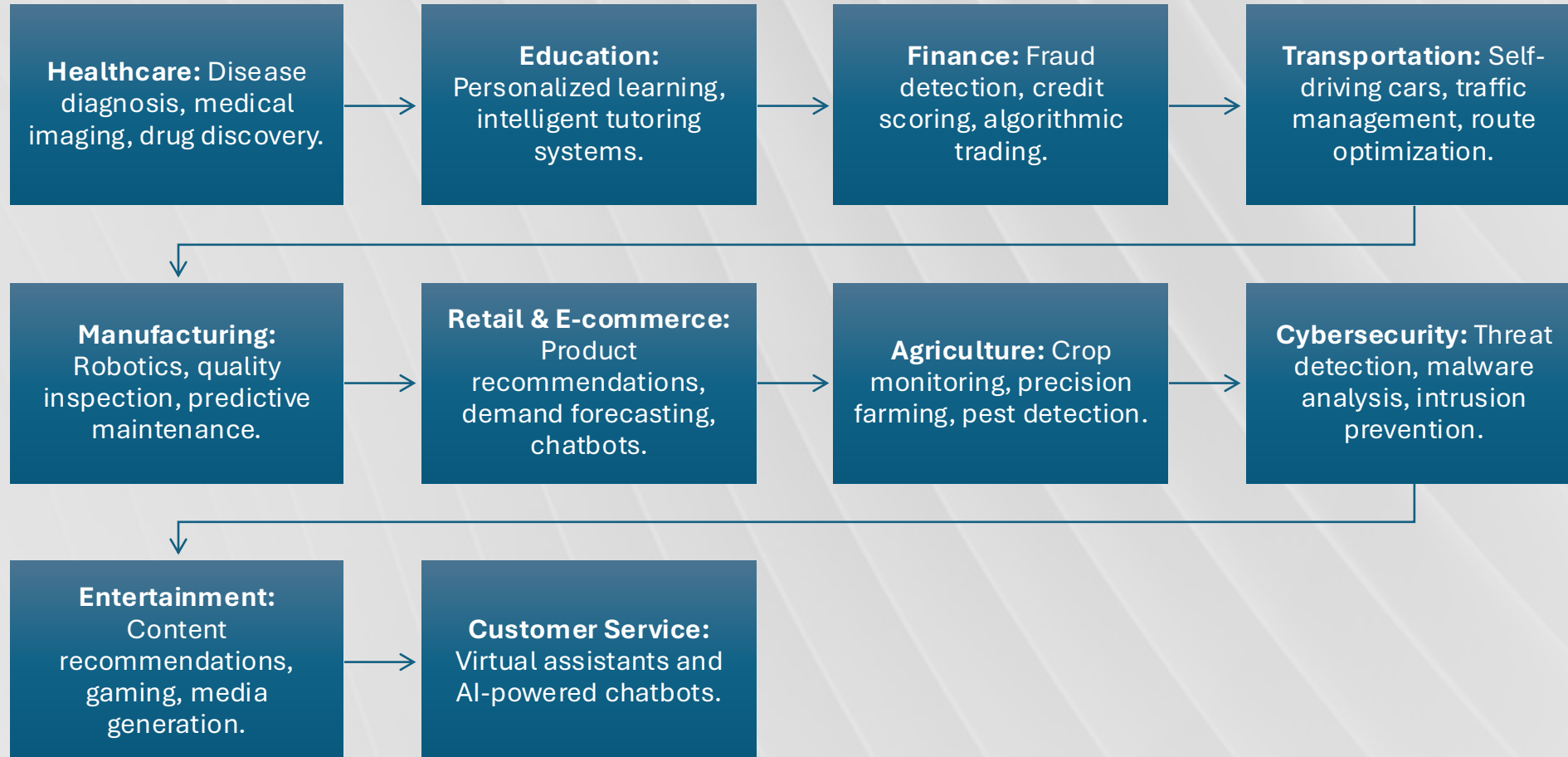
- **Learning:** Learns from data and improves performance over time.
- **Reasoning:** Makes logical decisions based on available information.
- **Problem Solving:** Finds efficient solutions to complex problems.
- **Knowledge Representation:** Stores and organizes information for decision-making.
- **Perception:** Understands inputs such as images, speech, and sensor data (face recognition, self-driving cars).
- **Natural Language Processing (NLP):** Understands and generates human language.
- **Decision Making:** Selects the best action based on analysis.
- **Adaptability:** Adjusts to new data and changing environments.
- **Automation:** Performs tasks with minimal human intervention.
- **Self-Correction:** Learns from mistakes to improve future performance.

Goals of A.I

Goals of Artificial Intelligence



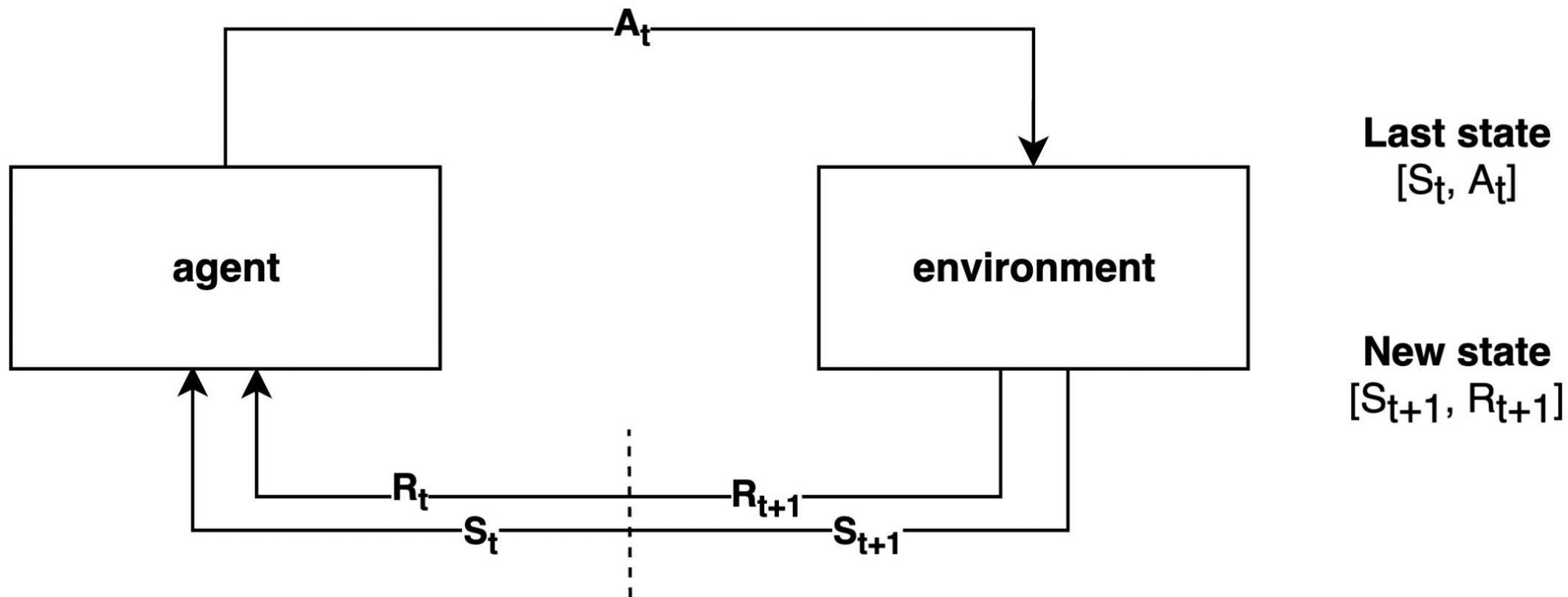
Applications of A.I



Agents and Environment

The A.I system consist of an agent and environment, **agents acts upon environment.**

- Agent:
- Agent is anything that can be viewed as perceiving its environment through **sensors** and acting upon that environment through **actuators**.
- Agents runs in the cycle of perceiving, thinking and acting.
- The agent receives information from the environment from the sensors and store and process the information in the agent program do the actions by using the **effectors**



A.I v/s Robot

AI

Programmed to think

Social Interaction

Learns



Robot

Programmed to do

Low level interaction

Only as smart as program

The A.I Problems

- This image lists various tasks that AI can perform, categorized into three groups: Mundane Tasks, Formal Tasks, and Expert Tasks. Here's a breakdown of each category:
- **1. Mundane Tasks:**
- These are **everyday tasks** that humans perform naturally but are challenging for AI due to their complexity and the requirement of a vast amount of contextual knowledge.
- **Perception:**
 - **Vision:** The ability to interpret and understand visual information from the world, such as recognizing objects, people, or scenes.
 - **Speech:** Understanding and processing spoken language.
- **Natural Language:**
 - **Understanding:** Comprehending the meaning of words, sentences, and texts.
 - **Generation:** Producing human-like text or speech.
 - **Translation:** Converting text or speech from one language to another.
- **Commonsense Reasoning:**
 - The ability to make judgments and decisions based on everyday knowledge that humans take for granted.
- **Robot Control:**
 - The ability to manage and direct the actions of a robot, particularly in dynamic environments.

Mundane Tasks

- Perception
 - Vision
 - Speech
- Natural language
 - Understanding
 - Generation
 - Translation
- Commonsense reasoning
- Robot control

Formal Tasks

- Games
 - Chess
 - Backgammon
 - Checkers -Go
- Mathematics
 - Geometry
 - Logic
 - Integral calculus
 - Proving properties of programs

Expert Tasks

- Engineering
 - Design
 - Fault finding
 - Manufacturing planning
- Scientific analysis
- Medical diagnosis
- Financial analysis

The A.I Problems

2. Formal Tasks:

- These are tasks that involve formal rules and structures, often in well-defined environments, and can include strategic or logical reasoning.
- **Games:**
 - **Chess:** A classic AI challenge due to the complexity and need for strategic thinking.
 - **Backgammon:** Another game where AI can be used to predict probabilities and make strategic decisions.
 - **Checkers:** Involves tactical planning and anticipating the opponent's moves.
 - **Go:** A complex board game known for its vast search space and the requirement for pattern recognition.
- **Mathematics:**
 - **Geometry:** Solving problems related to shapes, sizes, and the properties of space.
 - **Logic:** The study of reasoning and the principles of valid inference.
 - **Integral Calculus:** A branch of mathematics dealing with integrals and their properties.
 - **Proving Properties of Programs:** Verifying that software programs function as intended, particularly in terms of correctness and reliability.

The A.I Problems

3. Expert Tasks:

- These tasks require specialized knowledge and skills, often involving deep domain expertise.
- **Engineering:**
 - **Design:** Creating plans or specifications for constructing objects, systems, or processes.
 - **Fault Finding:** Identifying and diagnosing issues or defects in systems or products.
 - **Manufacturing Planning:** Strategizing and organizing the production process to optimize efficiency and output.
- **Scientific Analysis:**
 - Using AI to process and interpret scientific data, conduct experiments, or simulate complex systems.
- **Medical Diagnosis:**
 - AI assists in diagnosing diseases or medical conditions by analyzing patient data, symptoms, and medical history.
- **Financial Analysis:**
 - AI is used to analyze financial markets, assess risks, and make investment decisions based on vast amounts of data.

The A.I problems: Conclusion

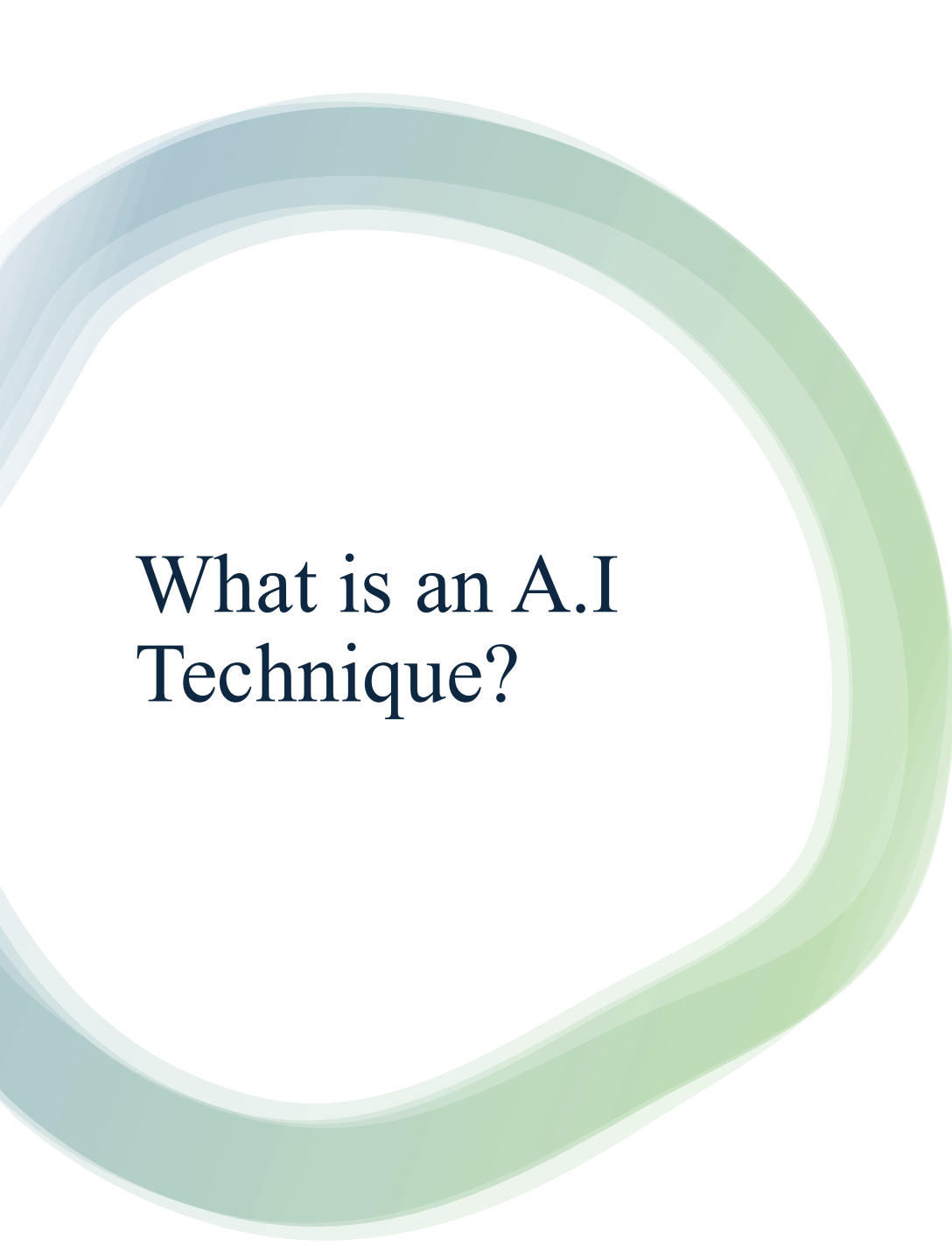
- This categorization highlights the broad range of applications for AI, from routine tasks that humans do naturally (but are complex for machines) to tasks requiring formal reasoning and specialized expert knowledge. The list provides a snapshot of the diverse challenges AI tackles across different domains, reflecting the interdisciplinary nature of the field.

The A.I Problems

Problems to answer

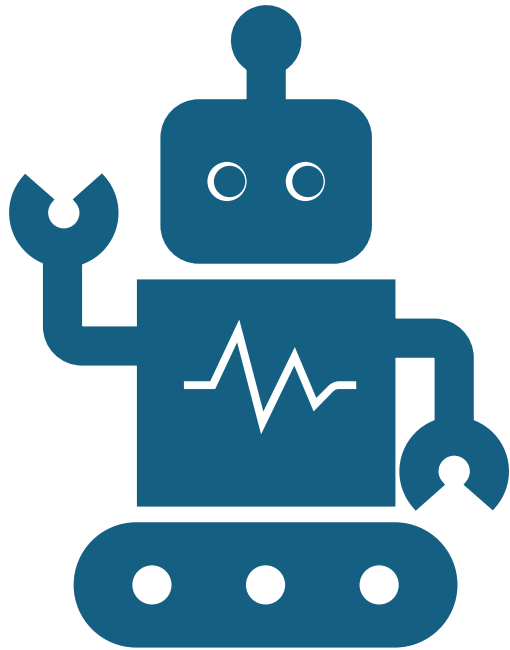
1. What are the underlying assumptions about intelligence?
2. What kinds of techniques will be useful for solving AI problems?
3. At what level of detail, if at all, are we trying to model Human Intelligence?
4. How will we know when we have succeeded in building an intelligent program?





What is an A.I Technique?

- An AI technique is a method that exploits knowledge that should be represented in such a way that:
 - The knowledge captures generalizations. it is not necessary to represent separately each individual situation, Instead, situations that share important properties are grouped together. If not having this property, inordinate amounts of memory and updating will be required, then it will be called “data” and not knowledge.
 - It can be understood by people who must provide it.[taking values automatically through several instruments] must provide knowledge in terms they understand.
 - It can easily be modified to correct errors and to reflect changes in the world.
 - It can be used in many situations even if its not totally accurate or complete.
 - It can be used to help overcome its own sheer bulk by helping to narrow the range of possibilities that must usually be considered.



Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn from data and improve their performance without being explicitly programmed. ML algorithms identify patterns in data and use them to make predictions or decisions.

Why Machine Learning is Important?

- **Automates Decision-Making** – Makes predictions and decisions without explicit programming.
- **Handles Large Volumes of Data** – Analyzes massive datasets quickly and efficiently.
- **Improves Accuracy** – Learns from data and becomes more accurate over time.
- **Identifies Hidden Patterns** – Discovers trends and relationships that humans may miss.
- **Enables Personalization** – Powers recommendation systems for shopping, entertainment, and social media.
- **Supports Real-Time Predictions** – Detects fraud, predicts demand, and monitors systems instantly.
- **Drives Innovation** – Forms the foundation of AI applications such as self-driving cars, healthcare, and robotics.
- **Reduces Time and Cost** – Automates repetitive tasks, improving productivity and efficiency.

Applications of ML



Email spam
filtering



Recommendation
systems



Fraud detection



Medical diagnosis



Image and speech
recognition

Neural Networks

A **Neural Network** is a machine learning model inspired by the structure and functioning of the **human brain**. It consists of interconnected processing units called **artificial neurons**, which work together to learn patterns from data and make predictions or decisions.

A Neural Network is a computational model made up of interconnected neurons that learns from data to recognize patterns, classify information, and make predictions.





Main Components

1. Input Layer

- First layer of the neural network.
- Receives the input data (images, text, numbers, etc.).
- Passes the data to the hidden layer(s).

2. Hidden Layer(s)

- One or more intermediate layers between the input and output.
- Extracts features and learns patterns from the input data.
- Performs computations using weights, biases, and activation functions.

3. Output Layer

- Final layer of the network.
- Produces the prediction or classification result.
- Number of neurons depends on the task (e.g., one neuron for binary classification, multiple neurons for multi-class classification).



Main Components

4. Neurons (Nodes)

- Basic processing units of the network.
- Receive inputs, perform calculations, and pass the output to the next layer.

5. Weights

- Numerical values that determine the importance of each input.
- Updated during training to improve the model's performance.

6. Bias

- An additional parameter added to the weighted input.
- Helps the network learn more effectively by shifting the activation function.

7. Activation Function

- Determines whether a neuron should be activated.
- Introduces non-linearity, enabling the network to learn complex patterns.
- Common activation functions: **ReLU, Sigmoid, Tanh, Softmax.**

How It Works



Each neuron performs calculations using **weights**, **biases**, and an **activation function**.



Information flows through one or more hidden layers.



The output is generated.



During training, the network adjusts its weights using **backpropagation** to reduce errors and improve accuracy.



Advantages of Neural Network

- **Learns Complex Patterns** – Can identify intricate relationships in large and complex datasets.
 - **High Accuracy** – Provides excellent performance for tasks such as image recognition, speech recognition, and classification.
 - **Automatic Feature Learning** – Learns important features directly from raw data, reducing the need for manual feature engineering.
 - **Handles Non-Linear Problems** – Models complex, non-linear relationships that traditional algorithms may struggle with.
 - **Adaptable and Scalable** – Improves as more training data becomes available and scales to large datasets.
 - **Robust to Noisy Data** – Can still perform well even when the input data contains some errors or noise.
 - **Supports Multiple Data Types** – Works with images, text, audio, video, and numerical data.
 - **Wide Range of Applications** – Used in healthcare, finance, autonomous vehicles, cybersecurity, Natural Language Processing (NLP), and recommendation systems.
-

Applications of Neural Network

Image and face
recognition

Speech
recognition

Natural Language
Processing
(ChatGPT,
translation)

Medical diagnosis

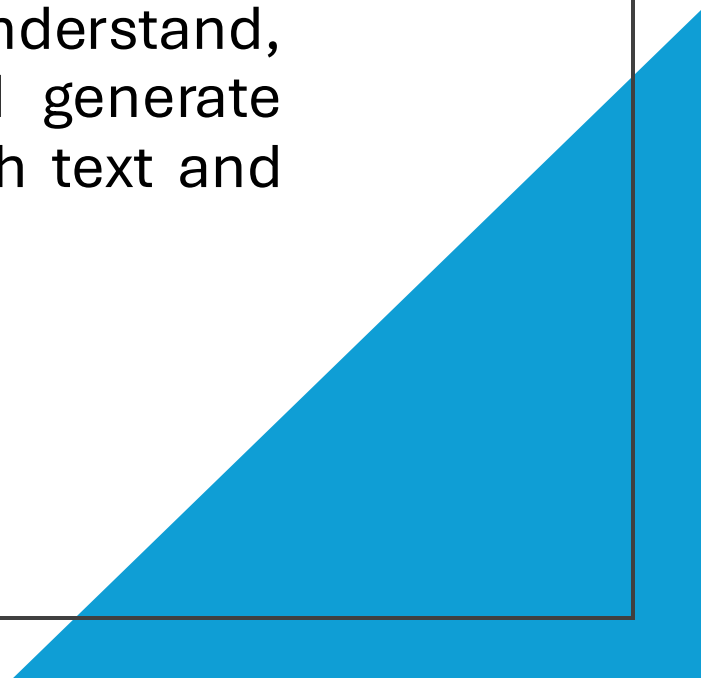
Fraud detection

Autonomous
vehicles

Recommendation
systems

Natural Language Processing (NLP)

Natural Language Processing (NLP) is a branch of **Artificial Intelligence (AI)** that enables computers to understand, interpret, process, and generate human language in both text and speech forms.



How NLP Works?



Text Input – Receive text or speech from the user.



Preprocessing – Tokenization, stop-word removal, stemming/lemmatization.



Feature Extraction – Convert text into numerical representations.



Model Processing – Use Machine Learning or Deep Learning models.



Output Generation – Produce predictions, translations, summaries, or responses.

Applications of NLP

- Chatbots and Virtual Assistants (e.g., ChatGPT, Siri, Alexa)
- Language Translation
- Email Spam Detection
- Sentiment Analysis
- Voice Recognition
- Search Engines
- Text Summarization
- Grammar and Spell Checking



NLP Tasks

1. Tokenization

- Splits text into smaller units called **tokens** (words or sentences).
- **Example:**
"I love AI" → ["I", "love", "AI"]

2. Stop Word Removal

- Removes common words that carry little meaning.
- **Example:**
"The cat is on the mat" → "cat
mat"

NLP Tasks

3. Stemming

- Reduces a word to its **root form** by removing prefixes or suffixes.
- The resulting word may not be a valid dictionary word.
- **Example:**
 - Playing → Play
 - Running → Run

4. Lemmatization

- Converts a word to its **base (dictionary) form**, called the **lemma**.
- Produces meaningful words using vocabulary and grammar rules.
- **Example:**
 - Running → Run
 - Better → Good
 - Studies → Study

NLP Tasks

5. Part-of-Speech (POS) Tagging

- Identifies the grammatical role of each word.
- **Example:**
"The cat runs" → The (Determiner), cat (Noun), runs (Verb)

6. Named Entity Recognition (NER)

- Identifies named entities such as people, places, organizations, and dates.
- **Example:**
"Sundar Pichai works at Google."
 - Sundar Pichai → Person
 - Google → Organization

NLP Tasks

7. Parsing

- Analyzes the grammatical structure of a sentence.
- Helps understand the relationship between words.
- Example: The little boy kicked the ball
- **The little boy** → Noun Phrase (Subject)
- **kicked** → Main Verb
- **the ball** → Noun Phrase (Object)

8. Sentiment Analysis

- Determines whether text expresses a positive, negative, or neutral sentiment.
- Example:
"The movie was amazing!" → Positive

NLP Tasks

9. Text Classification

- Assigns text to predefined categories.
- **Examples:** Spam detection, news classification, topic categorization

10. Text Summarization

- Generates a concise summary while preserving key information.